

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project

PYTHON

MYSQL

POWER BI



Made with GAMMA

Project Overview



Raw Dataset

3,900 rows



Python

Data cleaning & EDA



MySQL

SQL business analysis



Power BI

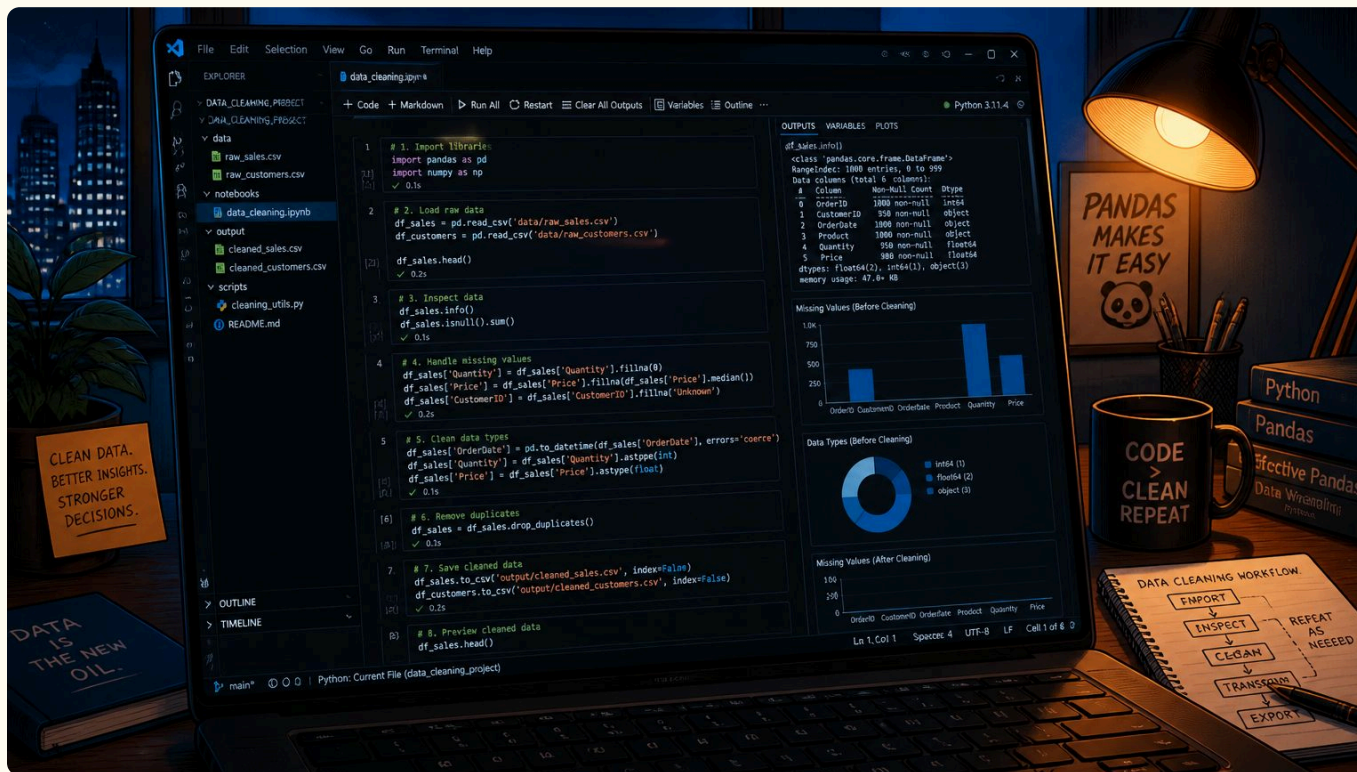
Interactive dashboard



Business Insights

Recommendations

Data Preparation — Python



Key Preprocessing Steps

- **Inspection:** Loaded dataset, checked dimensions, data types, and descriptive statistics
- **Quality:** Found 37 missing values in Review Rating; no duplicates
- **Cleaning:** Filled missing values by category median; standardized column names to snake_case
- **Feature Engineering:** Created Age Group segments; converted purchase frequency to numeric
- **Integration:** Connected cleaned dataset to MySQL for SQL-based analysis

SQL Business Analysis

Six key business questions answered through structured SQL queries.



Revenue by Gender

Male \$157,890 vs. Female \$75,191



Customer Segmentation

Loyal 3,116 · Returning 701 · New 83



Subscription Analysis

27% subscribers; 73% non-subscribers



Product Performance

Top 5 products by average review rating



Revenue by Age Group

Young Adults led at \$62,143



Shipping Analysis

Express avg \$60.48 vs. Standard \$58.46

Interactive Power BI Dashboard



Dashboard Features

- **KPI Cards:** 3.9K customers · \$59.76 avg purchase · 3.75 avg rating
- **Interactive Filters:** Subscription, Gender, Category, Shipping Type
- **Revenue Analysis:** Breakdown by category and age group
- **Customer Segmentation:** New, Returning, and Loyal visualizations
- **Product Insights:** Sales and revenue by product category

Key Business Insights

Clothing Dominates Revenue

Highest revenue and sales across all categories

Young Adults Lead Spending

Top revenue-contributing age group at \$62,143

Subscription Gap

Only 27% subscribed — significant growth opportunity

Loyal Customer Base

80% of customers classified as loyal

Express Shipping Premium

Slightly higher average spend (\$60.48 vs. \$58.46)

Top-Rated Products

Gloves, Sandals, and Boots rated highest — strong marketing potential

Business Recommendations

Increase Customer Subscriptions

Promote exclusive discounts and loyalty benefits to boost subscriber conversion.

Strengthen Customer Loyalty

Reward repeat customers with personalized offers to increase lifetime value.

Optimize Discount Strategy

Review campaigns regularly to drive sales while maintaining healthy margins.

Promote Top-Performing Products

Highlight best-selling, highly rated products to maximize engagement and conversions.

Focus on High-Value Segments

Target marketing toward highest-revenue age groups and loyal customers.

Project Summary

Tools Used

- Python (Pandas)
- MySQL
- Power BI

Skills Demonstrated

- Data Cleaning & EDA
- SQL Querying
- Dashboard Development
- Business Analysis & Data Storytelling

Transforming raw customer data into actionable business insights through analytics.